Patrick Foulis
CS455 Computer Networks
Lab 7
12 October, 2026

I. Your Forward Facing Table

```
Routing tables

Internet:
Destination        Gateway            Flags               Netif Expire
default            10.0.0.1           UGScg                 en0
10/24              link#11            UCS                   en0      !
10.0.0.1/32        link#11            UCS                   en0      !
10.0.0.1           94:6a:77:ce:44:f8  UHLWIir               en0   1175
10.0.0.2           e8:98:ee:91:eb:50  UHLWI                 en0   1152
10.0.0.25          32:8e:78:0:65:34   UHLWI                 en0    550
10.0.0.92/32       link#11            UCS                   en0      !
10.0.0.92          ea:17:d3:7f:8f:d4  UHLWI                 lo0
10.0.0.166         6a:f2:3e:60:c5:bf  UHLWIi                en0   1193
10.0.0.209         8e:2c:f7:70:ae:da  UHLWIi                en0   1198
10.0.0.250         98:17:3c:1f:ea:24  UHLWI                 en0    756
10.0.0.255         ff:ff:ff:ff:ff:ff  UHLWbI                en0      !
127                127.0.0.1          UCS                   lo0
127.0.0.1          127.0.0.1          UH                    lo0
169.254            link#11            UCS                   en0      !
224.0.0/4          link#11            UmCS                  en0      !
224.0.0.251        1:0:5e:0:0:fb      UHmLWI                en0
255.255.255.255/32 link#11            UCS                   en0      !
```

Row	Type / Interface
Default	Next Hop /en0
10/24	Connected /en0
10.0.0.1/32	Connected /en0
10.0.0.1	Next Hop /en0
10.0.0.2	Next Hop /en0
10.0.0.25	Next Hop /en0
10.0.0.92/32	Connected /en0
10.0.0.92	Connected /lo0

Row	Type / Interface
10.0.0.166	Next Hop /en0
10.0.0.209	Next Hop /en0
10.0.0.250	Next Hop /en0
10.0.0.255	Connected /en0
127	Connected /lo0
127.0.0.1	Connected /lo0
169.254	Connected /en0
224.0.0/4	Connected /en0
224.0.0.251	Connected /en0
225.225.225.225/32	Connected /en0

II. The Default Route

```
default            10.0.0.1           UGScg                  en0
```

My default gateway address is 10.0.0.1. The prefix 0.0.0.0/0 is used as a catch all because it matches any address because there is no number in-front of a 0 to match with.

III. Predictions Versus The Kernel

Destination	My Prediction	Kernels Answer	Agreement?
8.8.8.8	Next Hop	Next Hop	Yes it matched the default 0.0.0.0/0
10.0.0.1	Connected	Connected	Yes it matched the /24 subnet
10.0.0.50	Connected	Connected	Yes it matched the /24 subnet
127.0.0.1	Connected	Connected	Matches local loopback prefix.

IV. Longest Prefix Match In Binary

Prefix	Binary	Matching Leading Bits
0.0.0.0/0	00000000.00000000.00000000.00000000	0 bits
10.0.0.0/8	00001010.00000000.00000000.00000000	8 bits
10.20.0.0/16	00001010.00010100.00000000.00000000	16 bits
10.20.30.0/24	00001010.00010100.00011110.00000000	24 bits

Router R4 receives the packet because 10.20.30.0/24 has the longest matching prefix (24 bits) with the destination address 10.20.30.5.

There is no contradiction about describing an address four ways in a routing table because IP routing relies on the longest prefix match rule. Each entry represents a different level of network hierarchy. Allowing the router to choose what level of specificity to return.

V. Trace Route By Hand

I ran into an issue with this step. Initially when I ran it no matter how many times I incremented through the loop, it would skip TTL7 and after TTL8 would process nothing and just iterate through the loop. After consultation with AI I learned that it was because when you use Mac OS the -w increment is translated differently. Thus the time needed to be extended. The AI model gave me a new loop and this was its output.

```
~ % for n in $(seq 1 20); do
  printf "TTL %2d: " $n
  line=""
  for try in 1 2 3; do
    line=$(ping -c 1 -W 2000 -n -m $n 1.1.1.1 2>&1 | sed -n '2p')
    [ -n "$line" ] && break
  done
  echo "${line:-* (no reply)}"
  case "$line" in *"64 bytes from 1.1.1.1"*) break;; esac
done
TTL  1: 92 bytes from 10.128.1.1: Time to live exceeded
```

```
TTL  2: 36 bytes from 128.153.248.1: Time to live exceeded
TTL  3: 36 bytes from 199.109.109.85: Time to live exceeded
TTL  4: 76 bytes from 199.109.107.198: Time to live exceeded
TTL  5: 76 bytes from 199.109.107.206: Time to live exceeded
TTL  6: 36 bytes from 206.82.104.31: Time to live exceeded
TTL  7: * (no reply)
TTL  8: 36 bytes from 162.158.61.221: Time to live exceeded
TTL  9: 64 bytes from 1.1.1.1: icmp_seq=0 ttl=56 time=18.667 ms
```

The last line replies with an ICMP echo reply coming from our target 1.1.1.1. Where the rest of them returned with the time exceeded. This did not result in a TTL exceeded prompt because the TTL was within the bounds that were set .

VI. A Time Exceeded Packet

Sending Router: 10.128.1.1
Time To Live Value = 1
The router includes a copy of the original IP header and payload the receiving machine can link the ICMP error message to the exact outgoing packet and socket that triggered it.

VIII. The Tool

Trace route:

```
 1  10.128.1.1  5.758 ms  5.179 ms  10.263 ms
 2  128.153.248.1  8.487 ms  3.730 ms  52.872 ms
 3  199.109.109.85  19.176 ms  9.983 ms  8.718 ms
 4  199.109.107.198  11.740 ms  14.422 ms  14.668 ms
 5  199.109.107.206  17.840 ms  15.758 ms  17.128 ms
 6  206.82.104.31  55.893 ms  17.894 ms *
 7  * 199.27.132.1  79.584 ms *
 8  162.158.61.113  16.423 ms
    162.158.61.221  24.113 ms
    162.158.61.125  26.447 ms
 9  1.1.1.1  19.717 ms  19.875 ms  16.724 ms
```

By hand:

```
92 bytes from 10.128.1.1: Time to live exceeded
TTL  2: 36 bytes from 128.153.248.1: Time to live exceeded
TTL  3: 36 bytes from 199.109.109.85: Time to live exceeded
TTL  4: 76 bytes from 199.109.107.198: Time to live exceeded
TTL  5: 76 bytes from 199.109.107.206: Time to live exceeded
TTL  6: 36 bytes from 206.82.104.31: Time to live exceeded
TTL  7: * (no reply)
TTL  8: 36 bytes from 162.158.61.221: Time to live exceeded
TTL  9: 64 bytes from 1.1.1.1: icmp_seq=0 ttl=56 time=18.667 ms
```

They do in fact agree for the most part and for the major details. However there are a few differences, the first being the TTL 7 defined an IP address but there was a * * note. After some

research, I learned that this is because that exact probe timed out. The other difference is the 3 addresses in #8. This is because trace route allows multiple probes per ping. Whereas ping only allows one. But other than that it is the same.

IV. A Silent Hop

```
 % traceroute -n wikipedia.org
traceroute to wikipedia.org (208.80.154.224), 64 hops max, 40 byte
packets
 1  10.128.1.1  5.277 ms  3.443 ms  4.573 ms
 2  128.153.248.1  3.828 ms  5.238 ms  4.839 ms
 3  38.122.120.241  15.233 ms  11.905 ms  8.671 ms
 4  154.24.3.165  12.674 ms  10.359 ms  11.204 ms
 5  154.54.172.9  14.943 ms  14.174 ms  12.606 ms
 6  154.54.169.206  16.489 ms  15.903 ms  15.359 ms
 7  154.54.169.221  17.127 ms  19.234 ms  98.541 ms
 8  * * 154.54.9.26  23.109 ms
 9  62.115.140.180  27.240 ms
    62.115.140.27  26.559 ms  28.320 ms
10  62.115.138.58  24.715 ms
    62.115.122.202  27.927 ms  27.654 ms
11  62.115.139.35  28.982 ms
    62.115.136.201  28.946 ms
    62.115.139.35  30.243 ms
12  62.115.123.123  28.788 ms
    62.115.123.125  29.044 ms
    62.115.141.50  28.295 ms
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  *^C
```

One reason that the later addresses did not respond is because most network administrators configure firewalls at a perimeter of corporations or servers to drop icmp requests to prevent mapping of the network. The other reason that this could be happening is because some servers prevent icmp generation to protect their CPU resources. As being able to icmp a single server from multiple different machines, continuously can take down most servers in an attack. However it does continue to route the traffic

X. Two Paths

```
 % traceroute -n 1.1.1.1
traceroute -n 8.8.8.8
traceroute to 1.1.1.1 (1.1.1.1), 64 hops max, 40 byte packets
 1  10.128.1.1  95.752 ms  7.518 ms  4.428 ms
 2  128.153.248.1  4.573 ms  4.609 ms  4.476 ms
 3  199.109.109.85  10.015 ms  11.021 ms  10.113 ms
 4  199.109.107.198  10.859 ms  12.673 ms  12.218 ms
 5  199.109.107.206  16.921 ms  19.457 ms  17.488 ms
 6  206.82.104.31  14.779 ms  14.970 ms  15.332 ms
 7  * * 199.27.132.1  42.283 ms
 8  162.158.61.105  49.229 ms
    162.158.61.129  27.046 ms
    162.158.61.117  17.432 ms
 9  1.1.1.1  156.873 ms  39.998 ms  18.637 ms
traceroute to 8.8.8.8 (8.8.8.8), 64 hops max, 40 byte packets
 1  10.128.1.1  5.989 ms  4.920 ms  4.179 ms
 2  128.153.248.1  4.854 ms  4.082 ms  4.242 ms
 3  199.109.109.85  7.322 ms  9.461 ms  10.662 ms
 4  199.109.107.198  13.220 ms  11.981 ms  13.750 ms
 5  199.109.107.70  20.297 ms  21.225 ms  20.063 ms
 6  * * *
 7  * * *
 8  8.8.8.8  18.922 ms  21.592 ms  20.158 ms
```

199.109.109.85 is the first address which these two traces differ. The first 4 addresses that are routed to are the path out of the LAN. Thus some of those addresses can be safely assumed to be addresses within Clarkson's LAN.

XI. The AS Map

Hop Number	IP Address	AS Number	Organization Name
Hop 2	128.153.248.1	AS2637	Clarkson University
Hop 3	38.122.120.241	AS174	Cogent Communications
Hop 7	154.54.169.221	AS174	Cogent Communications
Hop 9	62.115.140.180	AS1299	Arelion (Telia Carrier)
Hop 12	62.115.141.50	AS1299	Arelion (Telia Carrier)

The traffic crossed three distinct autonomous systems denoted by the three AS numbers in the chart above. Boundaries on the internet are defined by these autonomous systems rather than the individual routers. The boundaries and decisions are made by the AS rather than the multiple routers that each AS has.

XII. TTL Interference

Host	Returned TTL	Inferred Initial TTL	Calculated Hop Count	Matches Trace?
1.1.1.1	56	64	8 hops 64 - 56 = 8	Yes
8.8.8.8	117	128	11 hops 128 - 117 = 11	Yes
wikipedia.org	49	64	15 hops 64 - 49 = 15	Yes
example.com	56	64	8 hops 64 - 56 = 8	N/A Did not trace

Any differences that do occur are because packets coming back to a machine often travel through a different set of routers and networks than the packets that were sent out. Also, we are only guessing the starting hop count based on standard computer defaults (like 64 or 128), which isn't always exact. But is good enough to work with.

XIII. The MTU Boundary

```
payload  1400: 1408 bytes from 1.1.1.1: icmp_seq=0 ttl=56 time=19.535
ms
payload  1470: 1478 bytes from 1.1.1.1: icmp_seq=0 ttl=56 time=19.037
ms
payload  1472: 1480 bytes from 1.1.1.1: icmp_seq=0 ttl=56 time=18.817
ms
payload  1473: PING 1.1.1.1 (1.1.1.1): 1473 data bytes
payload  1500: PING 1.1.1.1 (1.1.1.1): 1500 data bytes
```

The largest payload size was 1472 bytes. Thus the MTU = 1472 + 8 ICMP bytes + 20 IPv4 header bytes = 1500 bytes.

I saw a rejection from my own machine. There was no IP or ICMP reply from the server when it failed. Thus it never made it out of my own machine via rejection from the OS. If an

intermediate router drops an oversized packet because it cannot fragment it, and the ICMP "Destination Unreachable / Fragmentation Needed" message is filtered by a firewall, then the user's connection will hang. The client will retransmit the lost packet and it will continuously get dropped. It will eventually time out or be stuck indefinitely.

XIV. Translation

Yes the two addresses differ. My local address falls in the 10.0.0.0/8 while the public address falls in the 128.0.0.0 under Clarkson's network. On the way out the source IP address and the source port of were rewritten to the 128.0.0.0 address. On they way back in the 128.0.0.0 address was rewritten to my local 10.0.0.0 address on the LAN. The return path was made possible by the dynamic NAT table entry created by the outgoing packet. The router is performing address-and-port translation (NAPT) which allows traffic to get out under the public address and when it comes back as a private address.

A translating router relies on the NAPT table entry created by an outbound packet to map return traffic back to an internal machine. Because an unsolicited inbound packet from the Internet has no state table entry, the router has nothing telling it which private IP address should receive the packet, so it drops it automatically. Something like a zoom call needs to work around this, because the packet stream does not come from a specific corresponding outgoing packet. To make a local server reachable from the Internet, a Port Forwarding rule is required to allow an unsolicited incoming packet to be routed to a private address.

XV. Specification Questions

1) "The time is measured in units of seconds, but since every module that processes a datagram must decrease the TTL by at least one even if it process the datagram in less than a second, the TTL must be thought of only as an upper bound on the time a datagram may exist." The specification defines TTL to be in seconds, however the work in section 3 of this lab proved that my machine interprets TTL as a router hop counter. Which has nothing to do with the execution time.
2) "If this field contains the value zero, then the datagram must be destroyed." Matches TTL 1: 36 bytes from 10.128.1.1: Time to live exceeded
3) "The receiver of the fragments uses the identification field to ensure that fragments of different datagrams are not mixed." Without this ID number, a busy server receiving multiple split-up files at the same time wouldn't know which pieces belong to which file, mixing them up and corrupting the data.
4) "Internet Header + 64 bits of Original Data Datagram: The internet header plus the first 64 bits of the original datagram's data." Returning the IP header plus the first 8 bytes of original data gives the machine enough data to figure out where the command that sent the failed packet
5) To avoid the infinite regress of messages about messages etc., no ICMP messages are sent about ICMP messages." If this rule did not exist then if a packet containing an error had an error then the network would fall into an infinite loop.

XVI. Reflection

Hop by hop routing allows information to be sent without any single router knowing the entire map of every address on the internet. It saves a massive amount of data from needing to be stored if you can just increment closer and closer to the destination. It also makes the network scalable. Instead of updating the master map every time a new device is added to the internet. You can just know the proximity of other devices on the network and be connected. What is more difficult are things where the exact path needs to be known. It is hard to troubleshoot problem areas within a global scale because nothing is known about where the packet will be going. Which makes governing specific end to end connections extremely difficult. And global network engineering is hard because each router makes a specific independent decision.